Execise 2

Zhuge X.K.

January 6, 2026

Exercise 3.4

Exercise 3.6

Derive

$$\hat{G}^R(\tau) = -\frac{\mathrm{i}}{\hbar}\theta(\tau)\mathrm{e}^{-\frac{\mathrm{i}}{\hbar}E_n\tau} |\psi_n\rangle \langle\psi_n|$$

using the following expansion

$$\hat{G}^R(E) = \frac{1}{(E+\mathrm{i}\eta)\hat{I}-\hat{H}}\sum_n |\psi_n\rangle \langle\psi_n| = \frac{1}{E-E_n+\mathrm{i}\eta}\sum_n |\psi_n\rangle \langle\psi_n|$$

Execise 3.7

For 1D free electron

$$\left[E \pm \mathrm{i}\eta + \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\right] G^{R/A}(x, x'; E) = \delta(x - x')$$

Find two exact solution: $G_1(x - x', E) = G^R$ corresponding to "+", and $G_2(x - x', E) = G^A$ corresponding to "-".